Part 1: Python Fundamentals (1–15)

Beginner

Topic: Variables, Data Types, Operators, Conditions, Loops

1. Basic Calculator: Take two numbers and perform addition, subtraction, multiplication, division, floor division and modulus.

2. Temperature Converter: Convert Celsius to Fahrenheit and Fahrenheit to Celsius.

3. Even or Odd: Take an integer and determine whether it is even or odd.

4. Positive, Negative or Zero: Identify the type of a given number.

5. Largest of Three: Find the largest among three numbers without using `max()`.

6. Leap Year Checker: Determine whether a given year is a leap year.

7. Grade Calculator: Take marks and calculate grades (A+, A, A-, B, C, F).

8. Multiplication Table: Print the multiplication table of a given number.

9. Sum of Natural Numbers: Calculate the sum from 1 to N using a loop.

10. Factorial Calculator: Calculate N! using a loop.

11. Prime Number Checker: Determine whether a number is prime.

12. Fibonacci Series: Generate the first N Fibonacci numbers.

13. Reverse a Number: Reverse an integer without converting it to a string.

14. Palindrome Number: Check whether a number is a palindrome.

15. Armstrong Number: Check whether a number is an Armstrong number.

Part 2: Strings, Lists, Tuples & Dictionaries (16–25)

Beginner–Intermediate

Topic: Data Structures and String Manipulation

16. String Reversal: Reverse a string without using slicing (`[::-1]`).

17. Vowel Counter: Count vowels and consonants in a sentence.

18. Word Counter: Count the total number of words in a paragraph.

19. Character Frequency: Count how many times each character appears in a string.

20. Anagram Checker: Check whether two strings are anagrams.

21. List Statistics: Find the sum, average, minimum and maximum of a list without using built-in functions.

22. Remove Duplicates: Remove duplicate elements from a list while preserving order.

23. Second Largest Number: Find the second-largest unique number in a list without sorting.

24. List Sorting: Sort a list in ascending order without using `sort()` or `sorted()`.

25. Student Management System: Use a dictionary to store student names, IDs, marks and grades. Allow adding, searching, updating and deleting student records.

Part 3: Functions, Algorithms & Problem Solving (26–35)

Intermediate

Topic: Functions, Recursion, Searching, Sorting

26. Prime Numbers in a Range: Print all prime numbers between two given numbers.

27. GCD and LCM: Calculate the Greatest Common Divisor and Least Common Multiple of two numbers.

28. Recursive Factorial: Calculate factorial using recursion.

29. Recursive Fibonacci: Generate Fibonacci numbers using recursion.

30. Binary Search: Implement binary search on a sorted list.

31. Linear Search: Find an element and return its index using linear search.

32. Bubble Sort: Implement bubble sort manually.

33. Selection Sort: Implement selection sort manually.

34. Matrix Addition: Add two matrices using nested lists.

35. Matrix Multiplication: Multiply two matrices using nested loops.

Part 4: Python for Data Analysis & ML Preparation (36–50)

ML Preparation

Topic: NumPy, Pandas, Statistics, Data Processing

36. NumPy Array Creation: Create 1D, 2D and 3D arrays using NumPy.

37. Array Operations: Perform addition, subtraction, multiplication and division on two NumPy arrays.

38. Array Statistics: Calculate mean, median, standard deviation, variance, minimum and maximum using NumPy.

39. Array Indexing: Extract specific rows, columns and elements from a 2D NumPy array.

40. Reshape Arrays: Convert a 1D array into a 2D matrix using `reshape()`.

41. Random Data Generation: Generate 100 random numbers and calculate their statistical properties.

42. Pandas DataFrame: Create a DataFrame containing student names, IDs, ages and CGPA.

43. CSV File Handling: Read a CSV file using Pandas and display its first 10 rows.

44. Missing Values: Identify and handle missing values using `isnull()`, `dropna()` and `fillna()`.

45. Data Filtering: Filter students whose CGPA is greater than 3.50.

46. Data Grouping: Group student records by department and calculate average CGPA.

47. Data Visualization: Create a bar chart, histogram and scatter plot using Matplotlib.

48. Correlation Analysis: Generate a dataset of study hours and exam marks. Calculate their correlation using Pandas.

49. Outlier Detection: Detect outliers in a numerical dataset using the IQR method.

50. Mini ML Dataset Project: Create a student performance dataset with at least 100 records and perform:

- Data collection and CSV storage.
- Data cleaning.
- Missing-value handling.
- Exploratory Data Analysis (EDA).
- Statistical analysis.
- Data visualization.
- Correlation analysis.
- Train-test split preparation using Scikit-learn.